

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (MCA)

Semester – II University Examination May 2010

CA707 Relational Database Management System - I

Date: 19.05.10 ,Wednesday

Time: 12:30 p.m to 3:30 p.m

Maximum Marks:70

Instructions:

1. Answer both sections in the separate answer books.
2. Figures to the right indicate full marks allocated to question.
3. Assure suitable data wherever necessary.

SECTION-I

Q1 (a) Define the following terms.

1. Entity 2. Integrity 3. Redundancy 4. Concurrency

04

(b) Consider the relational database schemas.

07

emp(empno,ename,job,mgr,sal,hiredate,comm,deptno)

dept(deptno,dname,loc)

where the primary keys are underlined?

Write SQL statements for each of the following queries

1. List the department number and number of employees in each department.
2. List total salary, maximum salary, minimum salary, and average salary of each department
3. Display the name of employee's whose salary between 2000 and 5000.
4. List employee details in ascending order of salary.
5. List the name of employees whose name have exactly 5 characters.
6. List the different jobs available in the emp table.
7. List the name of employees whose name does not contain 'S'

Q2 (a) Discuss the different approaches of database design.

04

(b) Construct an E-R diagram of a book-store. Assume necessary attributes and constraints. Use generalization – specialization. The book-store has various warehouses, each with a unique code, address and phone number. Each warehouse stocks books. Details of books such as book no, year of publishing, title, price. Details of publishers like id, name, address, phone etc. Details of authors like name, address, phone, authorized etc are maintained. A customer can place his order for books and depending on availability will be delivered the books. Convert it into Relational Schema.

06

(c) What are the limitations of File systems?

02

OR

- Q2 (a) Explain ER model with example. 06
- (b) Draw an Entity-Relationship diagram for Student Information System which shows the personal and academic details of all the students of an institute enrolled in different departments for different courses. 04
- (c) List the components of DBMS. Explain each in brief. 02
- Q3 (a) What is normalization? Why Normalization is required? Explain 1NF, 2NF and 3NF with example. 06
- (b) What is Data anomaly? Explain different types of anomalies in Database. 04
- (c) Define functional Dependency. Give Example 02
- OR
- Q3 (a) Discuss the problems with 3NF. Explain with Example 4NF and BCNF. 06
- (b) Explain the Non-Loss (Lossless) Decomposition with Example. 04
- (c) Distinguish between Parallel database and Object-oriented database. 02

SECTION - II

- Q4 (a) Explain in detail the 3-level architecture of data models. 05
- (b) What are the advantages and disadvantages of SQL? 04
- (c) Distinguish between Object-oriented database and Distributed database. 02
- Q5 (a) What is Relational Algebra? Explain native operators of relational algebra with example. 08
- (b) Discuss Relational calculus in detail 04
- OR
- Q5 (a) Discuss the set-theory relational algebra operators with example 08
- (b) State the difference between Relational Algebra and Relational Calculus. 04
- Q6 (a) What is Query Optimization? Why it is required? Discuss the External Sorting Algorithm with example. 07
- (b) What are query trees? How are query trees built? 03
- (c) What kind of information is stored in the Database Catalog for evaluating the cost functions for cost estimation? 02
- OR
- Q6 (a) What is Query Processing? Explain Query Processor in Detail. 07
- (b) What are the different cost components required in estimating the cost of query optimization? 03
- (c) List different types of sorting algorithms. 02